TCS3351 CRYPTOGRAPHY AND DATA SECURITY
TUTORIAL 2
MATHEMATICAL BACKGROUND – I

1. a) Which of the following relations are true and which are false?

   (i) $5 \mid 26$ (ii) $3 \mid 123$ (iii) $27 \nmid 127$ (iv) $15 \nmid 21$

   Can 5 divide 26?==> No. => false

   3 | 123 = 41 true

   $27 \nmid 127$ true

   $15 \nmid 21$ true

   b) Find the results of the following operations.

   (i) -78 mod 13 (ii) 0 mod 13 (iii) 150 mod 15 (iv) -66 mod 33

   13 x -8 = -78, -78 mod 13 = 0

   0 mod 13 = 0

   150 mod 15 = 0

   -66 mod 33 = 0

2. a) Using the Euclidean algorithm, find the greatest common divisor of the pairs on integers, 88 and 220.

   b) Using the extended Euclidean algorithm, find the greatest common divisor of the pairs (4,7) and the value of s and t.

   Gcd(88,220) =
   88/2 = 44
   44/2 = 22
   22/2 = 11
   88 = 2 x 2 x 2 x 11
   220/2 = 110
   110/2 = 55
   55/5 = 11
   220 = 2 x 2 x 5 x 11

   GCD = 2x2x11 = 44

   Gcd(7,4)

   7 = 4.1 + 3
   4 = 3.1 + 1
   3 = 3.1 + 0

   Gcd(7,4) = 1

   Gcd(220,88)
   220 = 88.2 + 44 (last reminder)
   88 = 44.2 + 0

   GCD = 44

3. Perform the following operations using reduction first.

(i) (4223 +17323) mod 10
(ii) (2468 – 43) mod 15
(iii) (144 ×34) mod 12
(iv) (221 × 23) mod 22

(4223 +17323) mod 10
= 4223 mod 10 + 17323 mod 10
= 3 mod 10 + 3 mod 10
= 6

(2468 – 43) mod 15
= 2468 mod 15 – 43 mod 15
= 8 – 13
= -5
=-5 mod 15 = 10

(144 ×34) mod 12
= 144 mod 12 x 34 mod 12
= 0 x 34 mod 12
= 0

(221 x 23) mod 22
=221 mod 22 x 23 mod 22
= 1 x 1
=1

4. Check whether the multiplicative inverse exists for each of the following integers and if exists, find it using the extended Euclidean algorithm.

(i) 61 in $\mathbf{Z}_{90}$ (ii) 132 in $\mathbf{Z}_{180}$

multiplicative inverse exists when gcd(x,y) = 1

$\mathbf{Z}_{90}$={0,1,2,3,...,89}

Gcd(61, 90) = 1
Ext Euclidean algo
90 = 61. 1 + 29
61 = 29.2 + 3
29 = 9.3 + 2
9 = 4.2 + 1
4 = 1.4 + 0

Gcd(132, 180)
180 = 132.1 + 48
132 = 48.2 + 36

48 = 36.1 + 12
36 = 12.3 + 0
Gcd(132, 180) = 12 != 1 thus multiplicative inverse does not exist

5. Let us assign numeric values to the uppercase alphabet (A=0, B=1, …, Z=25). Answer the following using modulo 26.

   (i) (A+N) mod 26   (ii) (A+6) mod 26

   (ii) (Y-5) mod 26   (iv) (C-10) mod 26

(A+N) mod 26
= (0 + N)mod 26
= n mod 26
= N

(A+6) mod 26 = 6

(Y-5) mod 26
= 24 – 5 mod 26 = 19 (t)

(iv) (C-10) mod 26 = (2 – 10) mod 26 = -8
-8 mod 26 = 18 (S)

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

References:

1) Behrouz A. Forouzan, Debdeep Mukhopadhyay, "Cryptography and Network Security", Third Edition, McGraw-Hill, 2015.

2) William Stallings, “Cryptography and Network Security – Principles and Practice”, Seventh Edition, Prentice Hall, 2017.

Extra Exercises:

1.9. Use the Euclidean algorithm to compute the following greatest common divisors.
(a) $\gcd(291, 252)$.
(b) $\gcd(16261, 85652)$.
(c) $\gcd(139024789, 93278890)$.
(d) $\gcd(16534528044, 8332745927)$.

1.17. Do the following modular computations. In each case, fill in the box with an integer between 0 and $m-1$, where m is the modulus.
(a) $347 + 513 \equiv \boxed{} \pmod{763}$.
(b) $3274 + 1238 + 7231 + 6437 \equiv \boxed{} \pmod{9254}$.
(c) $153 \cdot 287 \equiv \boxed{} \pmod{353}$.
(d) $357 \cdot 862 \cdot 193 \equiv \boxed{} \pmod{943}$.
(e) $5327 \cdot 6135 \cdot 7139 \cdot 2187 \cdot 5219 \cdot 1873 \equiv \boxed{} \pmod{8157}$.
(*Hint.* After each multiplication, reduce modulo 8157 before doing the next multiplication.)
(f) $137^2 \equiv \boxed{} \pmod{327}$.
(g) $373^6 \equiv \boxed{} \pmod{581}$.
(h) $23^3 \cdot 19^5 \cdot 11^4 \equiv \boxed{} \pmod{97}$.

1.18. Find all values of x between 0 and $m-1$ that are solutions of the following congruences. (*Hint.* If you can't figure out a clever way to find the solution(s), you can just substitute each value $x = 1$, $x = 2, \ldots$, $x = m-1$ and see which ones work.)
(a) $x + 17 \equiv 23 \pmod{37}$.
(b) $x + 42 \equiv 19 \pmod{51}$.

1.32. For each of the following primes p and numbers a, compute $a^{-1} \bmod p$
(a) $p = 47$ and $a = 11$.
(b) $p = 587$ and $a = 345$.
(c) $p = 104801$ and $a = 78467$.

References:
1) Jeffrey Hoffstein, Jill Pipher and Joseph H. Silverman, "An Introduction to Mathematica; Cryptography", Second Edition, Springer, 2014.